GolfBot: An Autonomous Rover for Divot Detection, Localization, and Repair on Turf Surfaces

Teodor Donchev

A conversion thesis submitted for the degree of
Master of Science

Maynooth University
Faculty of Science and Engineering
Electronic Engineering Department

August 2025

Head of the Department: Prof. Gerard Lacey
Supervisor: Prof. Gerard Lacey

Abstract

Maintaining high-quality golf courses requires regular divot repairs, which is a slow, repetitive, and tiring job for maintenance workers. The GolfBot project proposes an autonomous robot to identify and fix divots on golf courses, addressing the need for a more efficient and automated solution in turf management.

This project, GolfBot, is an autonomous vehicle built upon the Wumpus platform, enhanced with specialized hardware and software to perform its task. The robot uses advanced computer vision techniques, including a custom-trained object detection model for real-time divot identification and a 3D depth camera for accurately measuring divot dimensions. For navigation, it is equipped with RTK GPS modules for precise positioning. The mechanical repair mechanism consists of a hopper with a motor-screw mechanism for dispensing a sand-seed mixture, and rubber brushes for uniform spreading. The entire system is controlled using ROS2, with components developed in Python and C++.

The result of this project is a functional prototype of an autonomous divot-repairing robot. This thesis documents the design, implementation, and evaluation of the GolfBot, demonstrating its capability to improve the quality and efficiency of golf course maintenance.

Declaration

Acknowledgments

Contents

Abstract **i**

Declaration **ii**

Acknowledgments **iii**

1 Introduction **1**

1.1 Motivation 1

1.2 The GolfBot System 1

1.3 Thesis Outline 1

2 Literature Review **3**

2.1 Autonomous Robots in Turf Maintenance 3

2.2 Divot Detection and Analysis 3

2.3 Automated Repair Mechanisms 3

2.4 Summary 3

3 Design **4**

3.1 Overall Design 4

3.1.1 Hardware Architecture 4

3.1.2 Software Architecture 4

4 Implementation **5**

4.1 Computer Vision System 6

4.1.1 Introduction/Purpose 6

4.1.2 Data Set Collection and Preprocessing 6

4.1.3 Model Training 6

4.1.4 Hardware 6

4.1.5 Software 6

4.1.6 Implementation in ROS 6

4.2 Navigation System 6

4.2.1 Introduction/Purpose 6

4.2.2 Hardware Used 6

4.2.3 Software Implementation 6

4.3 Mechanical Dispenser 6

4.3.1 Introduction/Purpose 6

4.3.2 Hardware Design and 3D-Printed Components 6

4.3.3 Software Integration in ROS 6

4.4 Graphical User Interface (GUI) 6

4.4.1 Introduction/Purpose 6

4.4.2 Software and Framework 6

4.4.3 Functions, Features, and Control 6

5 Experimental Results **7**

5.1 Computer Vision System Evaluation 8

5.1.1 Divot Detection Accuracy 8

5.1.2 Real-World Performance 8

5.2 Navigation System Evaluation 8

5.2.1 Positional Accuracy 8
5.2.2 Path Following and Search Algorithm Performance 8
5.2.3 Limitations 8
5.3 Mechanical Dispenser Evaluation 8
5.3.1 Dispensing Accuracy and Consistency 8
5.4 GUI Evaluation 8
5.4.1 Usability and User Feedback 8
5.5 Overall Performance of the GolfBot Prototype 8

6 Future Work **9**
6.1 Model Enhancements 9
6.1.1 Expanding the Dataset 9
6.2 Hardware Enhancements 9
6.2.1 Enhanced Visual System 9
6.2.2 Improved Drivetrain 9
6.2.3 Upgraded Dispenser 9
6.2.4 Autonomous Recharging 9
6.2.5 Enhanced User Interface 9
6.2.6 Automated Refilling Stations 9

7 Conclusion **10**
7.1 Summary of Work 10
7.2 Contributions 10

A Name **11**

List of Figures

3.1 GolfBot System Diagram 4

List of Tables

Chapter 1

Introduction

You can cite like this [1] for number only or you can cite like this for the author: Di Carlo et al. [1].

1.1 Motivation

1.2 The GolfBot System

1.3 Thesis Outline

The following chapters will detail the project.

- **Chapter 2: Literature Review** will survey existing work on autonomous robots in agriculture and turf maintenance, divot detection techniques, and automated repair systems.
- **Chapter 3: Design** will present the overall design of the GolfBot, covering both the hardware and software architectures.
- **Chapter 4: Implementation** will provide a detailed account of the implementation of the GolfBot's main components: the computer vision system, the navigation system, the mechanical dispenser, and the graphical user interface.
- **Chapter 5: Experimental Results** will describe the experiments conducted to validate the performance of the GolfBot and present the results.

- **Chapter 6: Future Work** will discuss potential enhancements and future research directions for the GolfBot project.
- **Chapter 7: Conclusion** will summarize the project, highlighting its contributions and concluding remarks.

Chapter 2

Literature Review

2.1 Autonomous Robots in Turf Maintenance

2.2 Divot Detection and Analysis

2.3 Automated Repair Mechanisms

2.4 Summary

Chapter 3

Design

3.1 Overall Design

3.1.1 Hardware Architecture

3.1.2 Software Architecture

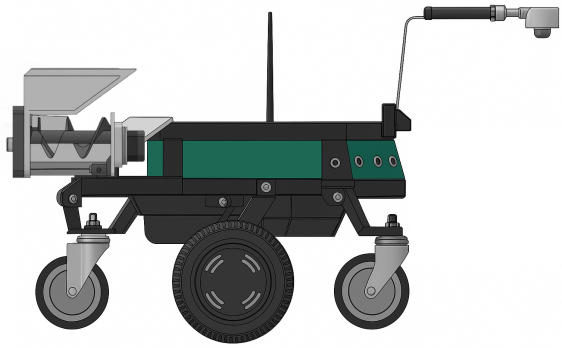

Figure 3.1: GolfBot System Diagram

Chapter 4

Implementation

4.1 Computer Vision System

4.1.1 Introduction/Purpose

4.1.2 Data Set Collection and Preprocessing

4.1.3 Model Training

4.1.4 Hardware

4.1.5 Software

4.1.6 Implementation in ROS

4.2 Navigation System

4.2.1 Introduction/Purpose

4.2.2 Hardware Used

4.2.3 Software Implementation

4.3 Mechanical Dispenser

4.3.1 Introduction/Purpose

Chapter 5

Experimental Results

5.1 Computer Vision System Evaluation

5.1.1 Divot Detection Accuracy

5.1.2 Real-World Performance

5.2 Navigation System Evaluation

5.2.1 Positional Accuracy

5.2.2 Path Following and Search Algorithm Performance

5.2.3 Limitations

5.3 Mechanical Dispenser Evaluation

5.3.1 Dispensing Accuracy and Consistency

5.4 GUI Evaluation

5.4.1 Usability and User Feedback

5.5 Overall Performance of the GolfBot Prototype

Chapter 6

Future Work

6.1 Model Enhancements

6.1.1 Expanding the Dataset

6.2 Hardware Enhancements

6.2.1 Enhanced Visual System

6.2.2 Improved Drivetrain

6.2.3 Upgraded Dispenser

6.2.4 Autonomous Recharging

6.2.5 Enhanced User Interface

6.2.6 Automated Refilling Stations

Chapter 7

Conclusion

7.1 Summary of Work

7.2 Contributions

Appendix A

Name

Bibliography

[1] J. Di Carlo, P. M. Wensing, B. Katz, G. Bledt, and S. Kim, "Dynamic Locomotion in the MIT Cheetah 3 Through Convex Model-Predictive Control," in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Oct. 2018, pp. 1–9.